Lecture 1: Class Overview and Background Review

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Welcome to MATH 1271!



Figure: Our Mascot, Maxwell, Deep in Thought

Today's Goals

1. Overview of the Course
2. Lecture on Algebra/Precalc Background
3. Worksheet Time!

Course Policies I

I've structured this course in the following manner:

1. We will start class with a *5 minute* quiz on the topic(s) covered in the previous lecture
2. We will then begin an 75 minute lecture, with a 5-ish minute break about halfway through
3. After another short break I'll hand out worksheets where we'll get a chance to apply the concepts we learned during lecture

Course Policies II

After each lecture you will have some *homework* due at 11:59 PM the following day. It will:

1. Be graded on an *honest completion* basis
2. Be split up into certain “levels”
3. And will conclude with one “challenge problem” designed to help you all develop more confidence for quiz/exam environments

Course Policies III

We will also have *weekly engagement portfolios* which are short 100 word pieces of writing to get you to engage in some metacognition about how you are learning in this course.
These will be consistently due at 11:59 PM on Mondays.

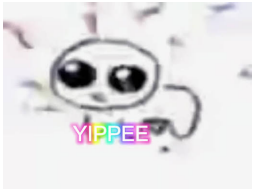
Course Policies IV

As mentioned before, we will have quizzes at the start of class relating to the content included in the previously-due homework, graded for accuracy.
We will also have *three* in class exams:

1. Exam 1: 10/6
2. Exam 2: 11/3
3. Exam 3: 12/8

Which will all take place during our regular class meeting here (Tate Hall 120).
Finally we will have a *cumulative* final exam on 12/17 from 12PM-3PM Location TBD.

Course Policies V



Some things that might make you happy!

- (At this point) your two lowest quizzes and two lowest engagement portfolios will be dropped.
- If you do poorly on an exam, you can have your performance on the final exam replace the grade if it is better.
- All this information is easily accessible and searchable on our syllabus on Canvas!

Office Hours?

I'm planning on holding two hours of office hours a week: in order that I hold them at the most opportune times please indicate your desires/availability on the When2Meet here.
There are also Office Hours that are held by some other TAs that you can find here.

Who Am I?

Hi! I'm Garrett! I am:

1. A 3rd Year PhD student in the math department
   (1) My research is in a field of math called *algebraic topology*
2. Someone who has TA'd for math courses three times (1x1271, 2x3283W) and taught once (1x2263)
3. The proud cat dad of the lovely, gracious, cute, and venerable Maxwell (who is our course banner on canvas!)
4. Super excited to teach you all some calculus this semester!!!

You can contact me at credi008@umn.edu

Who are you?

What is your:

1. Name
2. Major and standing (freshman, sophomore, etc)
3. Favorite *thing*?

Let's Do Some Math!

I want to spend today making sure we're all clear on the mathematical background necessary for success in this class:

1. Functions, their equations, graphs, and properties (e.g. domain and range)
2. Techniques with fractions
3. Trigonometric functions
4. Solving Equations

Functions I

What is a function?

Function

A function is an association from "things" of type A to other "things" of type B, typically denoted as $f : A \to B$.
For the purposes of our class, we will study **real-valued** functions of a **real variable** e.g. functions that take in a real number $x \in \mathbb{R}$ (read as x is in the real numbers) and output a real number $y \in \mathbb{R}$.

Equation of a Function

Real valued functions of a single real variable (we'll just call these functions from now on) can be **defined** by an equation which is just an algebraic specification of what the function does to an input number.

E.g. we can define the *squaring* function as $f(x) = x^2$. This means the value of the function f at some real number x is the square of x, so, for instance, $f(2) = 2^2 = 4, f(3) = 3^2 = 9$, etc.

Functions II

Graph of a function

The **graph** of a function $f : \mathbb{R} \to \mathbb{R}$ is a graphical/geometric way of interpreting what a function does.
To draw the graph, we first draw a *2D-Coordinate Grid*, and then draw all the points $(x, f(x))$ connected via a smooth line.

E.g.

Functions III

Sometimes functions behave differently depending on what kind of real number we input into them for instance the *absolute value function* leaves positive real numbers unchanged ($|2| = 2$) but turns negative numbers into their opposites ($|-2| = 2$). How can we define these functions?

Piecewise Function

A function is **defined piecewise** by the following method:

$$f(x) = \begin{cases} f_1(x), & \text{some condition on x} \\ f_2(x), & \text{some other condition on x} \\ f_3(x), & \text{maybe some other condition on x} \\ \vdots & \\ g(x), & \text{otherwise if none of the above conditions are met} \end{cases} \tag{1}$$

Functions IV

For example, we define the absolute value function as:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases} \tag{2}$$

and we can define some wacky function as:

$$wacky(x) = \begin{cases} x^2, & x > -5 \\ -x^3 + 2x + 1, & -5 \leq x < 2 \\ 1, & 2 \leq x < 3 \\ \frac{x}{2} + 3, & \text{otherwise} \end{cases} \tag{3}$$

Functions V

There are two important properties of a function that we need to talk about: domain and range.

Domain and Range of a Function

The domain of a function is the set of all values the function accepts as input. It is kind of arbitrary! However, we'll follow the convention that the domain should be the **largest** set of values it could accept as input.
Similarly, the range of a function is the set of all values the function could possibly output over all possible inputs. Given a specified domain, there is **exactly one** set of ranges so we don't have to deal with any arbitrariness.

For example, the *square-root* function $f(x) = \sqrt{x}$ would have:

- Domain: the *non-negative real numbers* $[0, \infty)$
- Range: again the *non-negative real numbers* $[0, \infty)$

What about a function like $f(x) = x^2 + 1$?

Fractions I

It's also worth going over some fraction foundations which will be very useful very soon in the course:

Addition of Fractions

Multiplication of Fractions

Fractions II

I also want to make clear what people mean by:

- Clearing out common factors/
- Factoring [blah] out/
- Multiplying by 1

The Multiplying by 1 Trick

Special Functions

We have a class of “special” functions:

- The exponential function $\exp(x) = e^x$
- It’s inverse, the natural logarithm $\ln(x)$
- The power functions x^n
- Their inverses $\sqrt[n]{x}$
- The trigonometric functions

Trigonometric Functions

Solving Equations I

Oftentimes we're interested in finding points where functions attain certain values or exhibit important behavior, how do we find them?
For instance, let's say we want to buy some ramen and the cost of ramen in dollars varies by weight in pounds according to the following function:

$$P(w) = x^2 + 7x.$$

If we only have 30 dollars, how much ramen can we buy?

Solving Equations II

How do we set that up mathematically?
We want to find the weight w so that:

$$P(w) = \$30 \tag{4}$$

So we can use our definition of P to get:

$$\begin{aligned} w^2 + 7w &= 30 \\ w^2 + 7w - 30 &= 0 \\ (w + 10)(w - 3) &= 0 \\ w = -10 \text{ or } w &= 3 \end{aligned}$$

And since -10 lbs of ramen doesn't make physical sense, we get that we can by 3 lbs of ramen with that much money.

Solving Equations III

Why did we choose to solve

$$(w + 10)(w - 3) = 0$$

for $w = -10, w = 3$
instead of solving

$$w(w + 7) = 30$$

for $w = 30$ or $w + 7 = 30$?

No Zero Divisors

If there are two real numbers a, b so that $ab = 0$, it **MUST** be the case that $a = 0$ or $b = 0$.

Question: If there are two real numbers a, b so that $ab = 30$, must it be the case that $a = 30$ or $b = 30$?

Solving Equations IV

How might we approach solving for x in an equation like:

$$(e^x)^2 - 5e^x + 6 = 0? \tag{5}$$

Or maybe like:

$$2\sin(x)^2 + \cos(x)^2 = 2? \tag{6}$$

Worksheet Time

Let’s get into groups and work on the worksheet!

End Of Class

That’s all folks!